Blind User Wearable Audio Assistance for Indoor Navigation Based on Visual Markers and Ultrasonic Obstacle Detection

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***Abstract* – This paper presents an indoor navigation wearable system based on visual markers recognition and ultrasonic obstacles perception used as an audio assistance for blind people. In this prototype, visual markers identify the points of interest in the environment; additionally this location status is enriched with information obtained in real time by other sensors. A map lists these points and indicates the distance and direction between closer points, building a virtual path. The blind users wear also glasses built with sensors like RGB camera, ultrasonic, magnetometer, gyroscope, and accelerometer enhancing the amount and quality of the available information. The user navigates freely in the prepared environment identifying the location markers. Based on the origin point information or the location point information and on the gyro sensor value the path to next marker (target) is calculated. To raise the perception of the environment, avoiding possible obstacles, it is used a couple of ultrasonic sensors. The audio assistance provided to the user makes use of an audio bank, with simple known instructions to indicate precisely the desired route and obstacles. Ten blind users tested and evaluated the system. The results showed rates of about 94.92% successful recognition of the markers using only 26 frames per second and 98.33% of ultrasonic obstacles perception disposed between 0.50 meters and 4.0 meters.**

I. Introduction

For blind people, walk freely is a challenge due to lack of information about the destination addresses, obstacles, etc. For them, there are plenty of new technologies that could be employed to decrease the difficulties caused by this impairment, making the relationship between man and environment more harmonious as possible [1].

Blind people use mainly the canes to move around and avoid obstacles. That is a very useful instrument and widely spread among blind people worldwide. Unfortunately, it is still a limited resource unable to provide an independent navigation and it cannot be used to detect objects or people more than a few feet away or above the waist of the user [1].

In fact, there are related proposals dealing with this problem based on modern technologies. Barathi and his associates presented a navigation technique using ultrasonic sensors on a cane and glasses to perceive obstacles on the ground level and above the head. They used standardized audio messages related to the perceived values to communicate with users [2]. Xiangxin and Mates described the building of a dog robot guide that uses ultrasonic sensors and an intelligent cord to communicate with users. The robot indicates through vibrations and jerks the action to be performed [3]. The robot also used a camera to capture images to recognize obstacles through classifiers Haar Cascade. Tian and his group proposed a tool that makes use of pre-estimated directions obtained by magnetic sensors, angular velocity calculus, gravity sensors, and RGB camera to guide their users. This work used Kalman filters to process data and to increase the location accuracy and the route calculations [4].

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Based on the already presented works and on the specific needs of the local blind community we decided to develop a new low cost tool able to help the indoor navigation through audio instructions. The guidance mode is done using visual markers arranged on the environment, linked as nodes in a bidirectional connected graph. Detection of obstacles is made using computer vision and ultrasonic perception.

II. Implementation of the Proposed System

The proposed wearable device is a combination composed of a hardware module and two software modules to deal with the collected environmental information. The software modules recognize visual markers through one RGB camera and static or dynamic obstacles through ultrasonic sensors at a safe distance. The users receive instructions through sound messages in their bone conduction headset [5]. Fig. 1 shows the proposed software architecture.

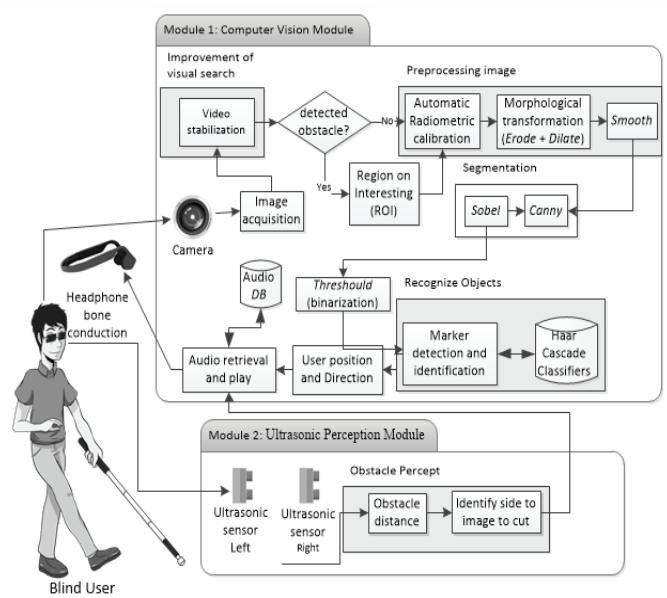

Fig. 1. Proposed interaction system architecture.

The hardware module consists in a pair of glasses modeled to receive a RGB camera and two ultrasonic sensors. A low cost mini PC was used to run the computer vision algorithms, store the Haar Cascade classifiers, store the audio database,

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control a bone conduction headset, and a camera. A microcontroller board controls the ultrasonic sensors, a magnetometer, a gyroscope, and one accelerometer. The initial prototype used a pair of ordinary glasses with a camera and ultrasonic sensors attached to them (Fig. 2).

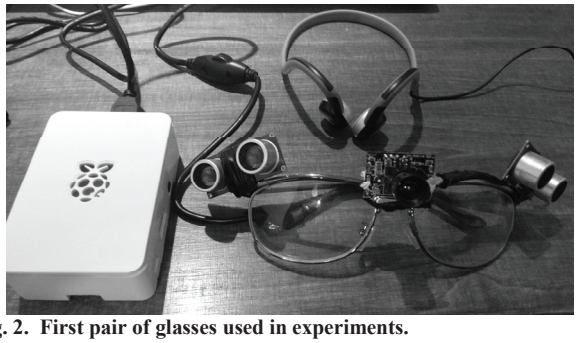

Fig. 2. First pair of glasses used in experiments.

After the first feedbacks about the usability of the prototype, we decided to design the glasses to hide the electronics and making it a more discreet device. A 3D printer was used to try out the users' suggestions. Some partial results are shown in Fig. 3.

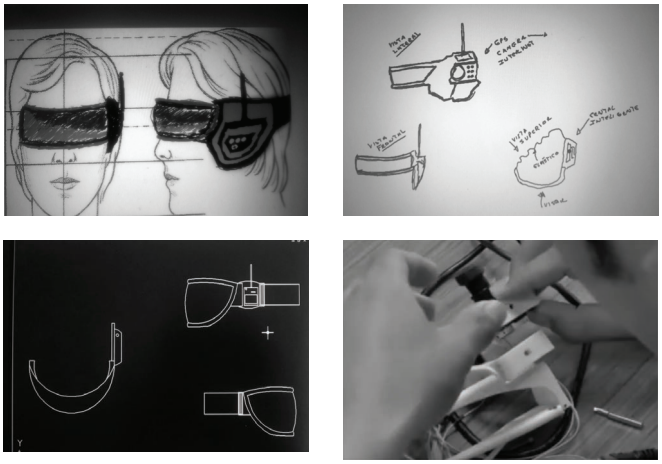

Fig. 3. Glasses prototype design process.

The software module consists in a Computer Vision module and the Ultrasonic Perception module operating in parallel mode. The computer vision module is used to recognize the visual markers and some obstacles placed in the environment. It first performs the video stabilization to correct the motion caused by the user walking. The image processing adapts the system to brightness variations and smooths the detected noises. The image segmentation reinforces the outer edges and the inner details of the objects before the recognizer step.

In step of image preprocessing, a radiometric calibration makes the system adaptable to brightness and luminous intensity changes. In order to increase the objects perception applies the following noise reducing techniques: histogram equalization, morphological transformation and Gaussian smoothing.

The image processing techniques has to two effects: remove important data or keep some noise. When removes important data, some objects lost characteristics that helps in identifying them. When leftover noise in the image, unconnected information or not belonging to any object harm the identification process.

After the image processing, it is performed the image segmentation step. This step consists in use filters to separate or group the pixels in patterns. In this project are used the Sobel [7] and Canny [6] filters respectively to highlight or rebuild the thin and thick edges of objects.

To build a Haar-Like classifier it is necessary to use two images sets: the positive, which contains the object one wants to map, and negative, which contains other objects. Haar-like is an automaton that searchs for a string of binarized data in a tree AdaBoost (tree where each node is a sub-tree).

OpenCV uses three algorithms to build the Haar-like cascade: Objectmarker, CreateSamples, and TrainCascade [9]. The Objectmarker creates a text file containing the image name and markers of the selected area. This file is converted into a vector through CreateSamples tool that standardizes the brightness, lighting and the images size for submission in the classification process. The Traincascade learns the pattern submitted in the vector and builds a Haar-Like tree.

Some related work suggest that to reach a good classifier it is necessary around 10,000 low quality images and a 14 level tree [8]-[9]. This project uses 2,000 high definition images and 18 level tree, empirically defined by tuning of the tree construction.

The indoor environments can be workplaces or homes. In these places have static objects (doors, walls, etc.), objects that can change places (furniture), and dynamic objects. The map of indoor environment is done using printed visual markers arranged at known points (static and furniture objects) [12].

The marker attributes like IDs, audio information, and relationship with other markers are recorded on a database. Each of the markers have a unique identifier and can be easily recognized by the system. Ultrasonic perception complete the system to trigger warning signals about the dynamic objects to the user and to the computer vision.

The markers identification process uses proximity methods and visual pattern analysis. The proximity method uses a symbolic and relative location of the marker, through abstract and statistical information. These markers positioning information are used in navigation.

During navigation, when the system detects a marker, the information about your position, direction to others markers, arrival time, etc. are updated and enhanced. The name of this method is navigation by proximity based on visual pattern. This method allows increasing the information about the states of the markes with the use. With an enough learning time, the system tends to stabilization and stops the modification of the data.

A. Guidance Mode

In order to run an experiment, six visual markers (A, B, C, D, E, and N) were adopted. The first goal was to find the best places to put the markers, considering the different height of users, ease of reading the markers. After these experiences, we decided to glue the markers on the floor. This decision helped to eliminate the need for users in turning their heads from side to side to detect the markers. These markers were placed in a corridor, as shown in Fig. 4.

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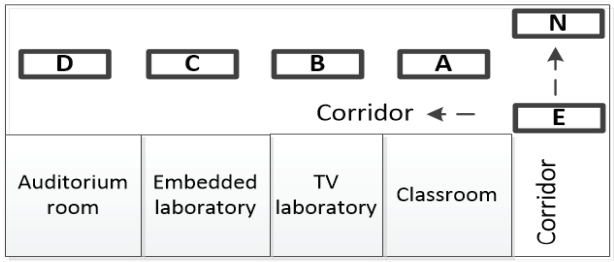

Fig. 4. Scenario for free mode navigation.

The visual markers are processed as nodes in a bidirectional connected graph. A weight value is given to each node using as criterion the distance between them. These values are recorded in an adjacency matrix (Table I). The weights 1, 2 and 10 indicate respectively the markers directly linked, route junction and indirectly linked.

TABLE I
ADJACENCY MATRIX FROM EXPERIMENTAL ENVIRONMENT

	A	B	C	D	E	N
A	10	1	10	10	2	10
B	1	10	10	10	10	10
C	10	1	10	10	10	10
D	10	10	1	10	10	10
E	2	10	10	10	10	2
N	10	10	10	10	2	10

For each marker, multiple sound instructions are associated. The user receives the accurate instruction, according of his/her trajectory. The system calculates the user direction and selects the audio instruction to assist the user.

For example, in a navigation from E (route junction) to the marker D, the system builds a trajectory and selects the audio instruction, as showed in Fig. 5.

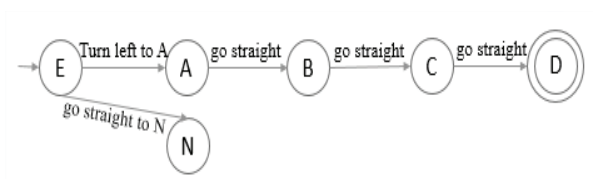

Fig. 5. Path between marker E and marker D with audio assistance.

Using the adjacency matrix the system can offer the best and shorter path and do corrections to get a specific marker if user takes a wrong way. In adjacency matrix, each cell have the actual marker weight value.

The system uses proximity algorithms to receive information about the symbolic location of markers and enhances this information with visual pattern analysis, through information on characteristics of the scene. The system increases the degree of accuracy with the use and performs fine adjustments to the predefined information of the markers.

III. EXPERIMENTS

Ten blind users evaluated the proposed navigation system efficiency. They received a brief explanation about the system. These users had the following profile: average height of 1.7 meters, average steps distance of 47 centimeters. The average of their speed walk was 20 steps in 17 seconds (1.17 steps in one second).

A path was chosen with a mandatory return to the starting point to be experienced by users. The path starts at an intersection (E) and follows a straight corridor to the bookmark D. At this point, the user indicated that he/ she would like to return to the initial marker to finalize his/ her navigation.

In our experiments, each user has done 5 navigations, totaling 50. In each navigation there were 6 markers covering a distance of 80 meters. The total number of read markers was 300 in all 50 navigations. Each navigation recorded the time spent in the full path and the delivery time of messages about the targets or about the obstacles in the path. The path itself was also registered on a database.

IV. RESULTS

In all navigations, users had contact with 300 markers and only 8 markers were missed or falsely identified due to light conditions. The success rates was 94.92% for markers detection and 98.33% to perception the obstacles. The average time to reach destination was 100 seconds.

In order to get subjective feedback about the system, users were subjected to six questions (Table 2). Question 1 was about the quality of the guidance obtained by the system. Question 2 was about the system capacity to provide freedom or independence to users in navigation along the path. Question 3 was about the extent to which the system was helpful in providing information about the current position and/or localization. Question 4 was about how much the system was reliable to the user. Question 5 was about the response time of the system and the Question 6 about the usability of the system.

TABLE II
RESULT QUESTIONS AFTER USERS EXPERIMENTS

		Performance Evaluation Level				
	Question	Excellent	Very Good	Good	Satisfactory	Poor
1	Guidance quality	55%	30%	5%	5%	5%
2	Independence	35%	35%	25%	5%	0%
3	Localization	35%	35%	25%	5%	0%
4	Reliability	55%	20%	10%	15%	0%
5	Response time	35%	35%	20%	10%	0%
6	Usability	5%	65%	20%	10%	0%

Looking at the two best results of performance evaluation (Excellent or Very Good), one realizes that: 85% approved the guidance qualify (question 1), 75% approved the reliability (question 4), 70% approved the system independence, model of localization adopted, response time and usability (questions 2, 3, 5 and 6).

The answers over these questions show clearly points where the research and prototype must be improved in order to satisfy the users.

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V. Evaluations

Two evaluation criteria were used: efficiency and time to recognizing a certain visual marker by classifiers. In order to evaluate efficiency in recognizing the classifiers, the proposed techniques presented by Rautaray and Agrawal [10], and by Wilson and Fernandez [11] were used. The algorithms were implemented so that the classifiers could be tested as originally proposed. In fact, they could be reproduced and parameterized according with descriptions given by the authors. Table III summarizes the results obtained from each marker classifier.

TABLE III
Adjacency matrix from experimental environment

	This work		[11]		[10]	
Marker	***True positive***	***False positive***	***True positive***	***False positive***	***True positive***	***False positive***
A	92,2%	7,8%	85,0%	7,0%	91,5%	8,5%
B	92,8%	7,2%	91,2%	8,8%	90,4%	7,6%
C	97%	3,0%	89,0%	5,0%	92,5%	7,5%
D	93,3%	6,7%	89,4%	10,6%	91,2%	8,8%
E	99,3%	0,7%	95,3%	4,7%	95,1%	1,9%

To evaluate the runtime of the computer vision system on the markers recognition process it was used as criterion the measurement of frames displayed every second. Fig. 6 contains the results.

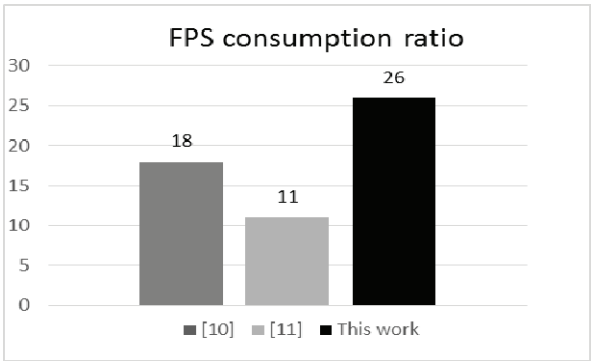

Fig. 6. Comparative diagram of the performance of related work.

It was noted that the work of the author [11] had a higher consumption. This work used only the classifier while others two adopt an association of classifiers to other algorithms, with low computational consumption.

VI. Conclusion and Future Work

This paper proposed the application of strategies that allowed to the user to navigate in an indoor environment through use of a wearable navigation system, discreet and with low cost. The system offered an audio assistance when recognized a visual marker on the environment and warned about potential obstacles unmapped, located on the route.

The development methodology of this work indicated that the Indoor Navigation could adopt a mapping of identification points by an adjacency matrix and processing in a self-organizing tree for offering guidance maps from these points.

A pair of glasses with camera and ultrasonic sensors was used as a wearable device for detection and identification of the points of interest through visual markers. The markers and obstacles mapped, when detected on the routes, were informed to the blind user through sound instructions and the obstacles unmapped by warning sound signals.

The results show that there are still many gaps to be treated for increased quality in indoor navigation directed to blind users and that the approaches used are promising. The next steps are more detailed studies on the indoor mapping techniques, definition of a language to be used in the identification of markers and run other experiments using different models of cameras, like the infrared camera, to increase the prototype confidence on your results, mainly when used it in low-light environments.

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